

OBJECT-ORIENTED MODELLING

Object-Oriented Programming

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Creating Models in Design ✓
- 2 Evolution of Programming Languages ✓
- 3 Four Design Principles
- 4 SOLID Principles



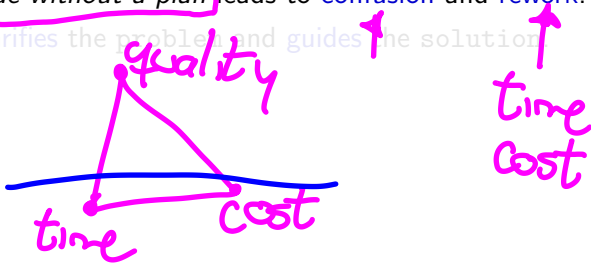
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Design Before Code

- **Design** should come before coding.
- Jumping into *code without a plan* leads to confusion and rework.
- Good design clarifies the problem and guides the solution.



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1

2



Understanding the Requirements

- **Requirements** must be well understood before design.
- Ask questions, clarify ambiguities, and document all requirements.
- Requirements define the scope and direction of the design.

↑
list ⇒ need



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business vocabulary



Design Based on the Problem

- **Design** should be driven by the problem, not by technology.
- Focus on what *needs* to be solved, *not just* how to implement it.
- Use the **problem statement** to identify *key objects* and their *relationships*.



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non-technical

system



Object-Oriented Approach

- The **object-oriented approach** models the **system** as a **collection** of interacting **objects**.
- Each **object** represents a **real-world** entity or concept.
- **Objects** encapsulate **data** and **behavior**.

objects ← *elements*
relations



Conceptual Design and Technical Design

- **Conceptual Design**: What the **system should do**, using high-level models.

- **Technical Design**: How the system will be implemented, using detailed diagrams and specifications.

- Both are essential for a successful software project.

Diagramas
Referencia Document

UML
Unified modeling language
draw.io



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USER

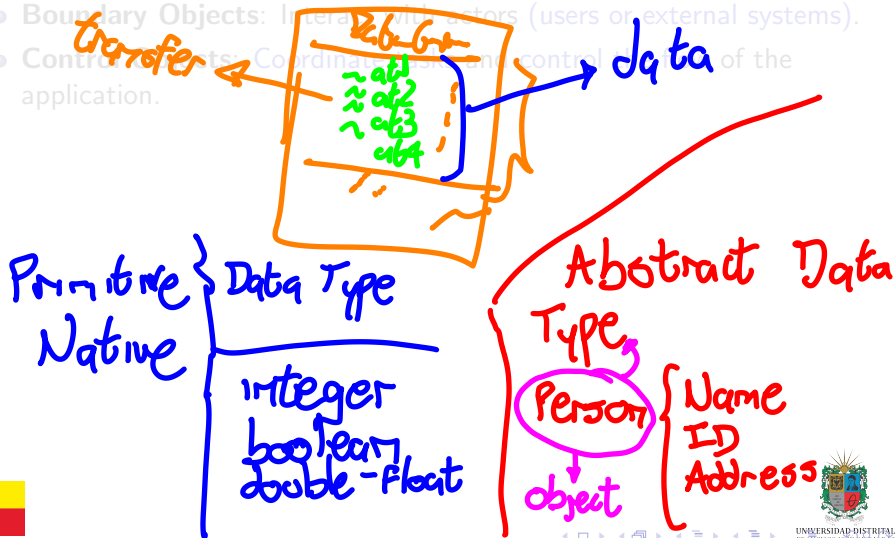
tech team

expectations



Categories of Objects

- **Entity Objects:** Represent information and data.
- **Boundary Objects:** Interact with actors (users or external systems).
- **Control Objects:** Coordinate tasks and control objects of the application.



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logic → most of the objects



Documentation in Software

- **Documentation** is essential for communication and maintenance.
- Includes requirements, design diagrams, user manuals, and code comments.
- Good **documentation** helps new team members *understand* the *system* quickly.



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old



solve problems

find a solution
innovative



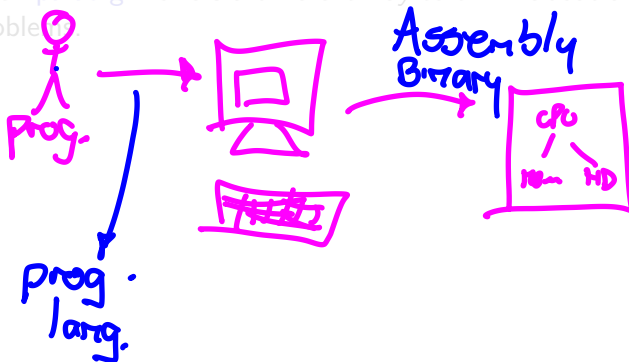
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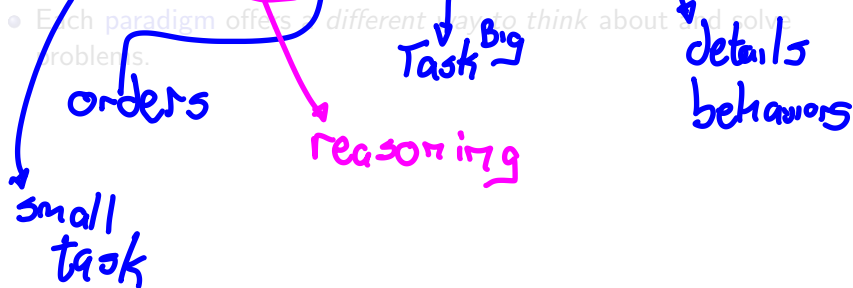
Talk with Machines: Programming Paradigms

- **Programming languages** are tools to **communicate with machines**.
- **Paradigms**: Imperative, Procedural, Object-Oriented, Functional, Logic
- Each paradigm offers a *different way to think about and solve problems*



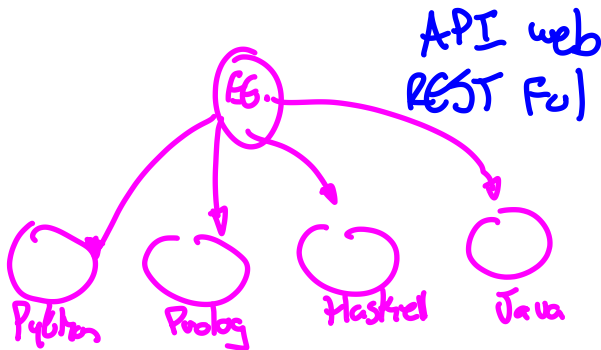
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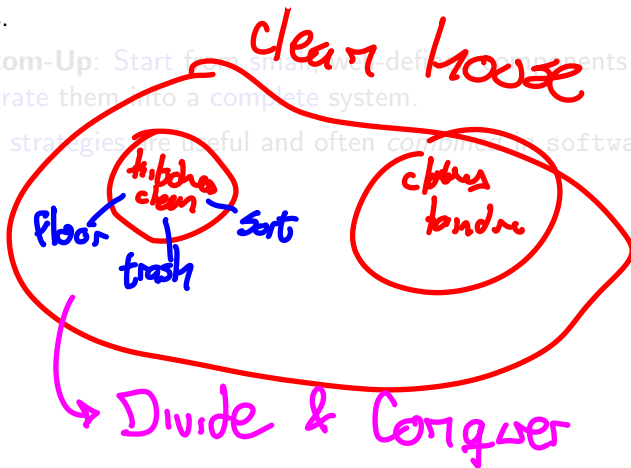
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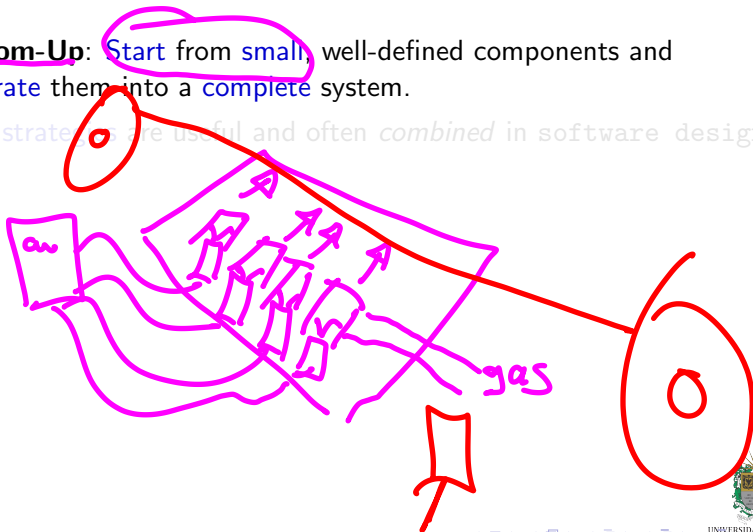
Strategies to Solve Problems

- **Top-Down:** Start from the **big** picture and **break** it down into smaller parts.
- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.
- Both strategies are useful and often **combined** in software design.



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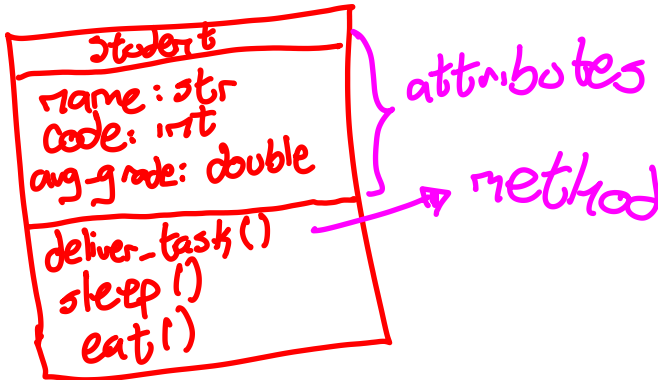
Object-Oriented Design and Contracts

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- **Contracts**: Define responsibilities and expectations between objects.
- **Contracts** help ensure correctness and robustness.



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UML Diagrams

- **UML** (*Unified Modeling Language*) is a standard way to visualize system design.

- Common diagrams: Class diagrams, Sequence diagrams, and Use case diagrams

- UML helps communicating design as clearly

34 C/205 46 Diagram
 Delay next support
 Activating
 use cases
 standard



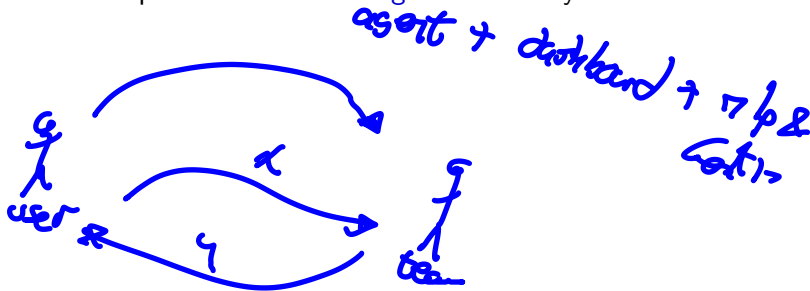
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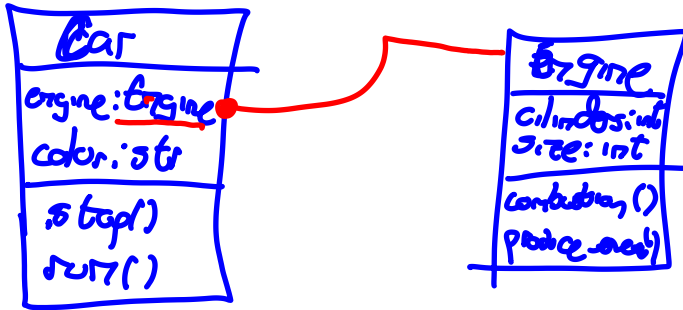
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Abstraction

- **Abstraction** means focusing on the essential features of an object.
- **Rule of Least Astonishment:** Design so users are not surprised by behavior.
- Consider context, basic attributes, and basic behaviors when *designing abstractions*.

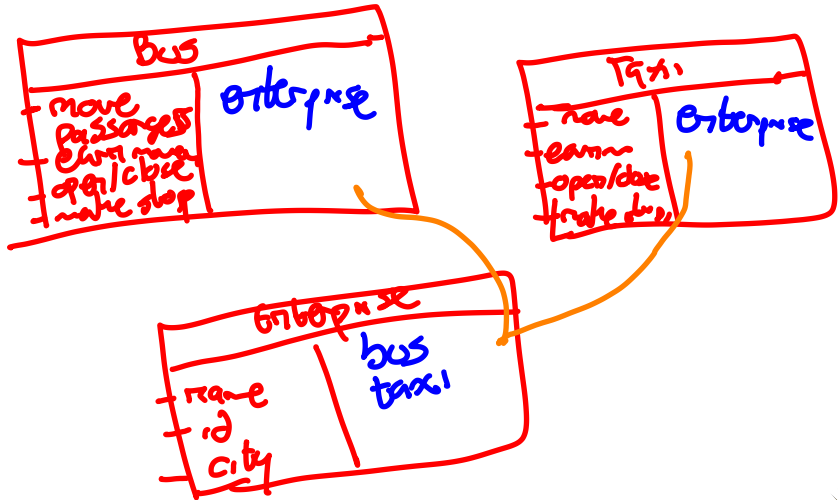


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Abstraction & CRC Cards



Encapsulation

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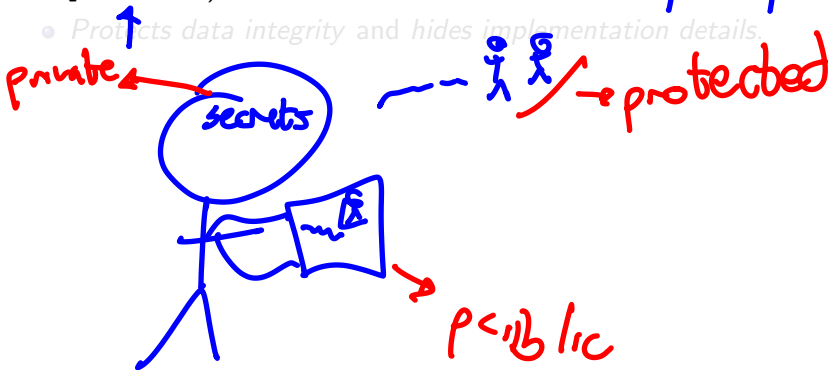
hide

information



Encapsulation

- **Encapsulation** bundles **attributes** and **methods** together.
- Expose only what is necessary (access levels: public, private, protected).
 - ↑
 - ↑
- *Protects data integrity and hides implementation details.*



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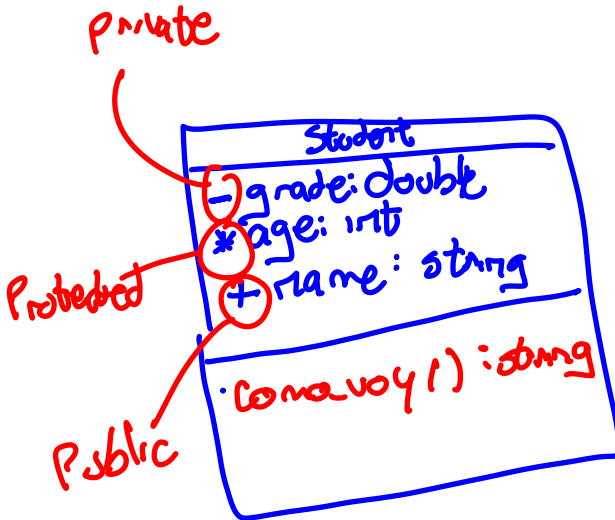
```

public class Student {
    private double grade;
    protected int age;

    public String comp-avg() {
        if (grade >= 3.0) {
            return "B+ or better";
        } else if (grade >= 2.0) {
            return "Satisfactory";
        } else {
            return "Needs more work";
        }
    }
}
  
```



Encapsulation & UML



Black Box Thinking

- **Objects communicate** through **well-defined interfaces**.
- **Rule of Least Knowledge**: Objects should **know** as **little as possible** about one another.
- **Black box**: Focus on what an object does, not how it does it.

Demo time!



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Data Integrity: Getters and Setters

- **Getters** and **Setters** are methods to access and modify object attributes.
- They help maintain **data integrity** by controlling how attributes are accessed and modified.
- Use them to enforce **validation rules** and **business logic**.

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Data Integrity: Getters and Setters

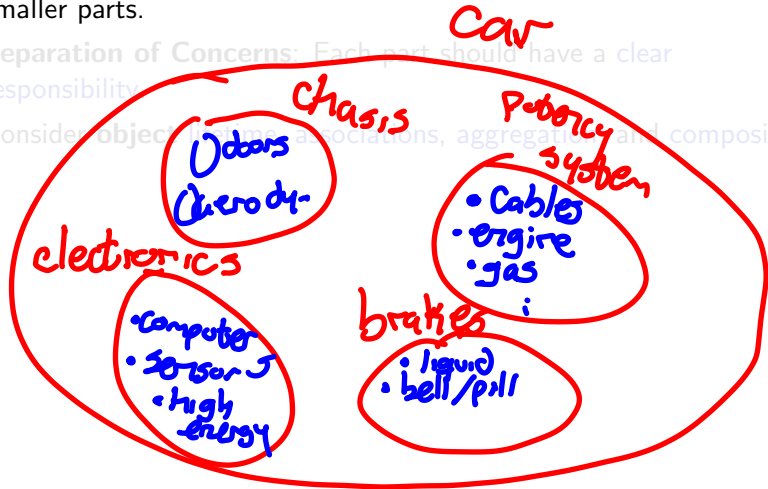
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- **Separation of Concerns:** Each part should have a clear responsibility.
- Consider object lifetime, associations, aggregation and composition.



Decomposition

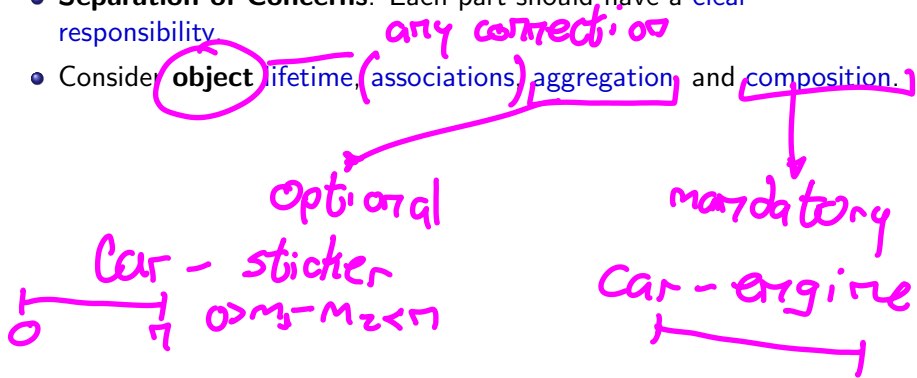
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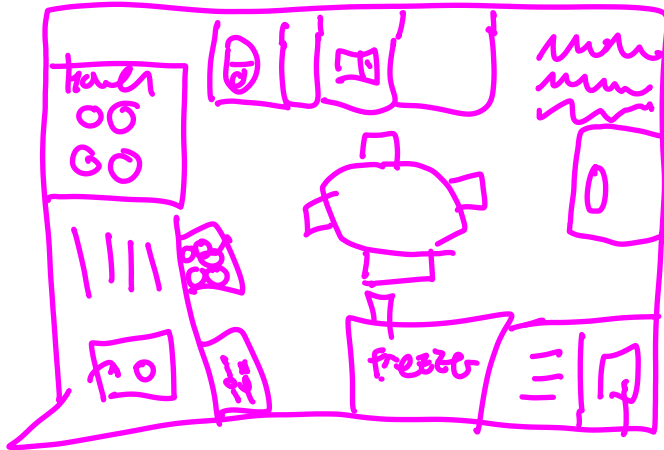


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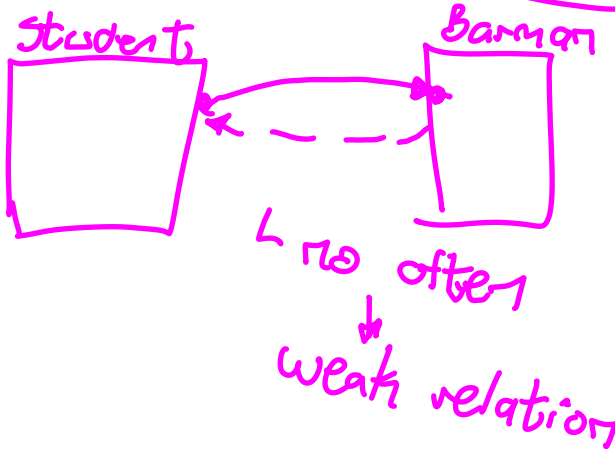


Decomposition Example: Kitchen in a House



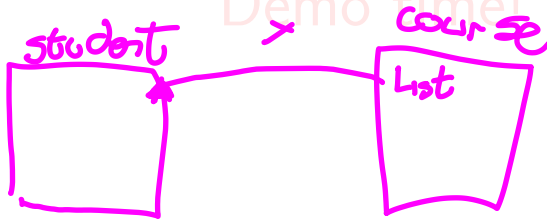
Association

A **relationship** between two classes where one class uses or interacts with another class.



Aggregation

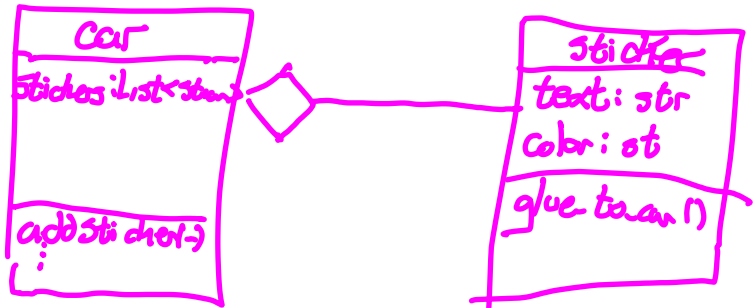
An **aggregation** is a **whole-part relationship** where one class is a **part** of another class, but **can exist independently**.



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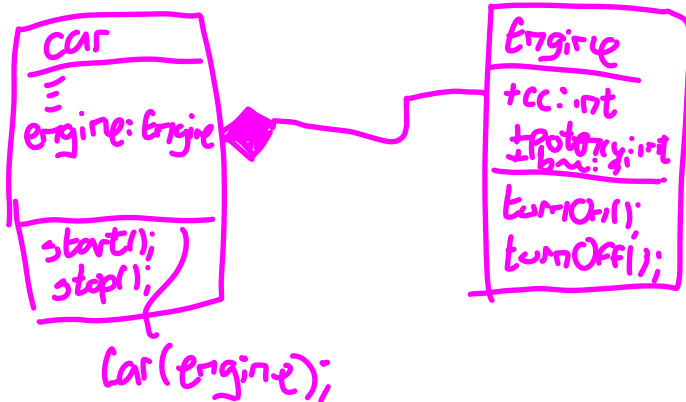
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Composition

Composition is a stronger whole-part relationship where one class is a part of another class and cannot exist independently.

Demo time!



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Demo time!



Generalization

- **Generalization** eliminates redundancy by extracting common features.

- **D.R.Y. Principle:** Don't Repeat Yourself.

- Behaviors can be generalized using inheritance, interface inheritance, and abstract classes.

- Polymorphism: Objects can be treated as instances of their parent class.

- Types of inheritance: single, multiple, interface-based.



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abstract
concrete

→ implemented logic

Fully abstract

transfer



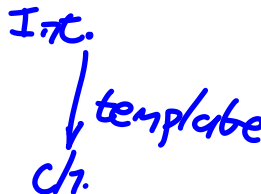
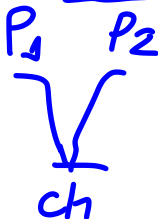
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Inheritance

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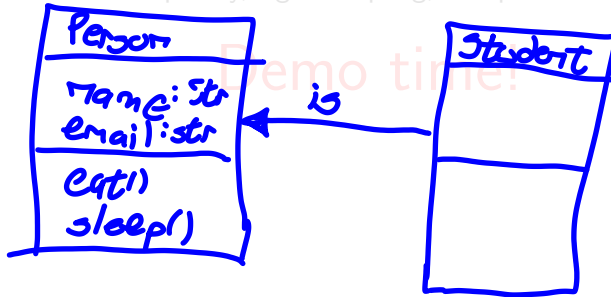
- **Base class:** The class being inherited from.
 - **Derived class:** The class that inherits from the base class.
 - **Benefits:** Code reusability, easier maintenance, and polymorphism.
 - **Drawbacks:** Complexity, tight coupling, and potential fragility.
- data + behaviour*

Demo time!



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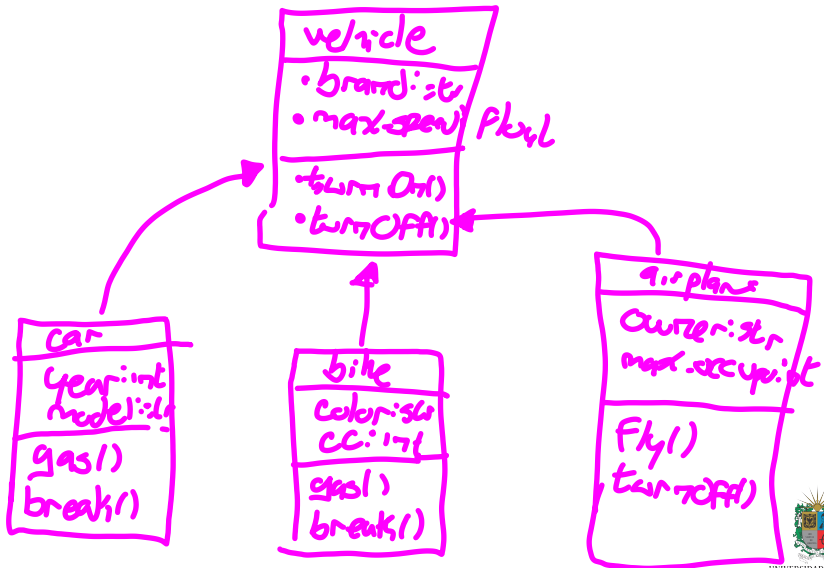
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- by inheritance*

Demo time!



Inheritance & UML



Interface Inheritance

- **Interface inheritance** allows a class to implement an interface without inheriting its implementation.
- Interfaces define a contract that classes must adhere to.
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abstract

↓
no logic
template

concrete

↓
logic

Demo time!



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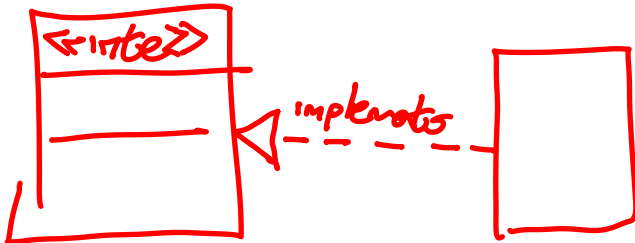
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Demo time!



Polymorphism by Inheritance

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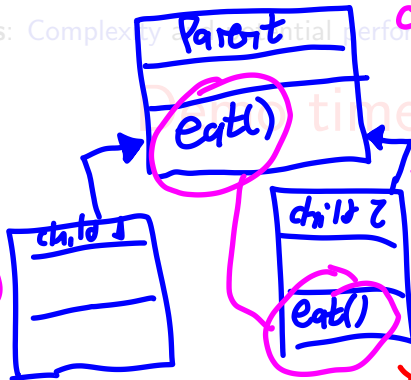
- **Drawbacks:** Complexity and potential performance issues.

multiple ways to make same action

child1.eat()

↓

"comiendo"



eat

eat

```

class Parent {
    public void eat() {
        print("comiendo");
    }
}
  
```

class Child2 {

@Override

```

    public void eat() {
        print("comiendo");
    }
}
  
```



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Demo time!



Polymorphism by Overloading

- **Polymorphism by overloading** allows multiple methods with the same name but different parameters.



```

class You {
    public void catch(int weight) {
        print("catch tennis of " + weight);
    }
    public void catch(double size) {
        print("catch football of size" + size);
    }
    public void catch(int size, int air) {
        print("soccer");
    }
}
  
```

Handwritten notes in pink and red ink:

- you.catch(5);* (with an arrow pointing to the `catch(int weight)` method)
- you.catch(2,5);* (with an arrow pointing to the `catch(int size, int air)` method)



Polymorphism by Overloading

- **Polymorphism by overloading** allows **multiple methods** with the **same name** but different parameters.
- **Benefits:** Improves readability and reduces complexity.
- **Drawbacks:** Can lead to confusion if not used carefully.

Demo time!



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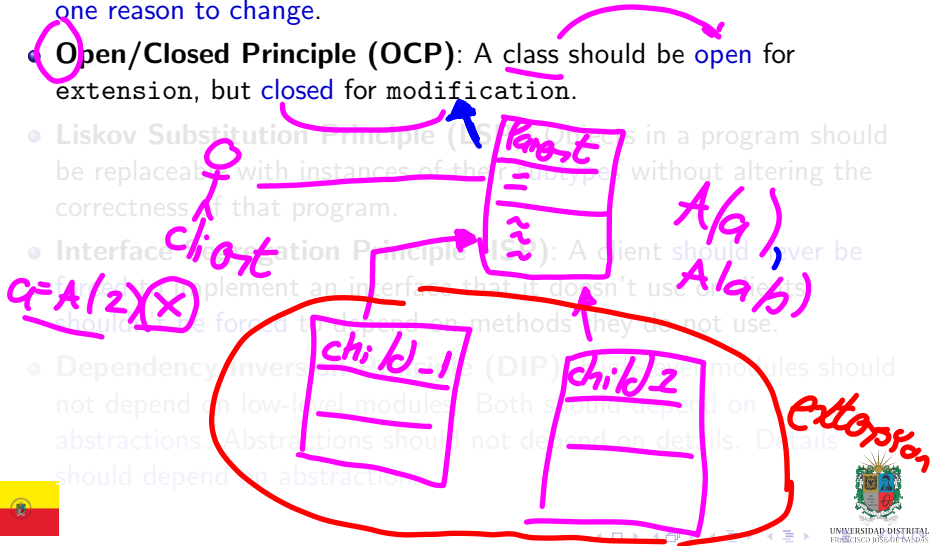


SOLID Principles

- **Single Responsibility Principle (SRP):** A class should have only one reason to change.
 - **Open/Closed Principle (OCP):** A class should be open for extension, but closed for modification.
 - **Liskov Substitution Principle (LSP):** Objects in a program should be replaceable with instances of their subclasses without altering the correctness of that program.
 - **Interface Segregation Principle (ISP):** A client should never be forced to implement an interface that it doesn't use or clients shouldn't be forced to depend on methods they do not use.
 - **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules. Both should depend on abstractions. Abstractions should not depend on details. Details should depend on abstractions.
- Handwritten notes:*
- Red circle around "A class" in SRP.
 - Red underline under "one reason to change" in SRP.
 - Red arrow pointing from "responsibility" to "activities" in the handwritten note.
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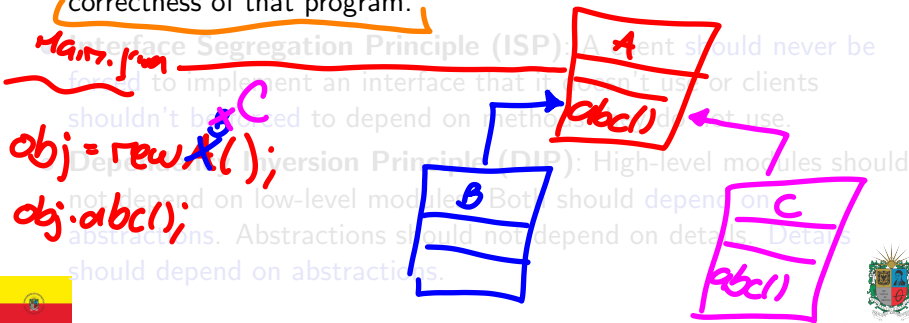


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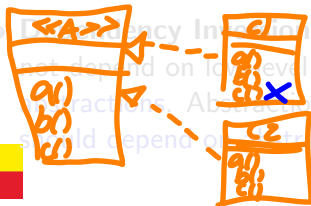
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Good Practices

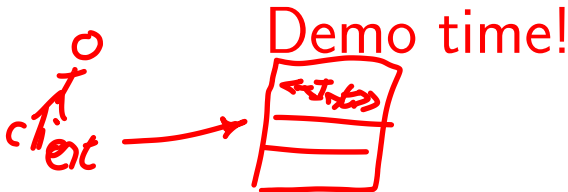
- **Composition over Inheritance**: Inheritance should be used only when there is a clear relationship between the base class and the derived class. In other cases, composition should be used. Inheritance is a powerful tool, but it is not always the best tool for the job. Inheritance is a way to achieve polymorphism, but it is not the only way to achieve polymorphism.
- **Code to Interfaces, not Implementations**: This principle is about designing your classes so that they depend on interfaces rather than concrete classes.

Demo time!



Good Practices

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Outline

- 1 Creating Models in Design
- 2 Evolution of Programming Languages
- 3 Four Design Principles
- 4 SOLID Principles



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/object-oriented-programming>

